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Which one of them reaches the bottom first? Explain your answer.

5. Write the relation between angular momentum and rotational kinetic energy. Draw a graph for the same. For two objects of same angular momentum, compare the moment of inertia using the graph.
6. A rectangle block rests on a horizontal table. A horizontal force is applied on the block at a height h above the table to move the block. Does the line of action of the normal force N exerted by the table on the block depend on h?
7. Three identical solid spheres move down through three inclined planes A, B and C all same dimensions. A is without friction, B is undergoing pure rolling and C is rolling with slipping. Compare the kinetic energies E_A, E_B and E_C at the bottom.
8. Give an example to show that the following statement is false. 'any two forces acting on a body can be combined into single force that would have same effect'.

V. Numerical Problems

1. A uniform disc of mass 100g has a diameter of 10 cm. Calculate the total energy of the disc when rolling along a horizontal table with a velocity of 20 cms^{-1}. (take the surface of table as reference)

Ans: 0.1028 J

2. A particle of mass 5 units is moving with a uniform speed of $v = 3\sqrt{2}$ units in the XOY plane along the line y = x + 4. Find the magnitude of angular momentum.

Ans: 60 units

3. A fly wheel rotates with a uniform angular acceleration. If its angular velocity increases from 20π rad/s to 40π rad/s in 10 seconds. Find the number of rotations in that period.

Ans: 150 rotations

4. A uniform rod of mass m and length ℓ makes a constant angle θ with an axis of rotation which passes through one end of the rod. Find the moment of inertia about this axis.

Ans: $\frac{1}{12} M\ell^2 \sin^2\theta$

5. Two particles P and Q of mass 1kg and 3 kg respectively start moving towards each other from rest under mutual attraction. What is the velocity of their center of mass?

Ans: Zero

6. Find the moment of inertia of a hydrogen molecule about an axis passing through its center of mass and perpendicular to the inter-atomic axis. Given: mass of hydrogen atom 1.7×10^{-27} kg and inter atomic distance is equal to 4×10^{-10}m.

Ans: 1.36×10^{-46} kg m^2

7. On the edge of a wall, we build a brick tower that only holds because of the bricks' own weight. Our goal is to build a stable tower whose overhang d is greater than the length ℓ of a single brick. What is the minimum number of bricks you need?

(**Hint:** Find the center of mass for each brick and add.)